Submitted by

Abhishek Gupta

Institute Name - Skill Cricle

NETFLIX

Email_id - abhi653210@gmail.com

Overview

This dataset provides a comprehensive overview of Netflix's content library, covering both movies and TV shows available on the platform. It includes key attributes such as title, type of content, director, cast, country of production, release year, rating, duration, and listed genres. The dataset helps in understanding content distribution across different countries, genres, and time periods. It also enables analysis of trends such as the growth of Netflix content over the years, popularity of specific ratings, and dominance of movies versus TV shows. Overall, this dataset is useful for content analysis, business insights, and data visualization projects related to the entertainment and streaming industry.

OBJECTIVE OF THE PROJECT

- To analyze the overall structure and characteristics of the Netflix dataset.
- To understand the distribution of movies and TV shows available on the platform.
- To identify trends in content release over different years.
- To examine popular genres, ratings, and content durations.
- To analyze country-wise contribution to Netflix content.
- To gain meaningful insights that can support business decisions and content strategy.

TOOLS & TECHNOLOGIES USED

- Pthyan
- Pandas
- Seaborn
- Matplotlib
- Google Collab
- Canva (for presentation)

DATA EXPLORATION

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```python
nf = pd.read_csv('/content/netflix_titles (1).csv')
nf
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...
...	...	...	...	...	...	...	...	...	...	...	...	...
8802	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007	R	158 min	Cult Movies, Dramas, Thrillers	A political cartoonist, a crime reporter and a...
8803	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2018	TV-Y7	2 Seasons	Kids' TV, Korean TV Shows, TV Comedies	While living alone in a spooky town, a young g...
8804	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	2009	R	88 min	Comedies, Horror Movies	Looking to survive in a world taken over by zo...
8805	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006	PG	88 min	Children & Family Movies, Comedies	Dragged from civilian life, a former superhero...
8806	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015	TV-14	111 min	Dramas, International Movies, Music & Musicals	A scrappy but poor boy worms his way into a ty...

8807 rows × 12 columns

1. **The dataset contains 8,807 rows and 12 columns, indicating a large and diverse Netflix content library.**
2. **It includes both Movies and TV Shows, with movies appearing more frequently than TV shows.**
3. **Key attributes such as release year, rating, duration, country, and genre allow multi-dimensional analysis.**
4. **Several columns like director, cast, and country contain missing values, showing real-world data quality issues.**
5. **The dataset supports trend analysis related to content growth over time, popular genres, and regional contributions, making it suitable for data analysis and visualization projects.**

DATAFRAME SUMMARY

```
nf.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

- The DataFrame contains 8,807 records and 12 columns, representing Netflix movies and TV shows.
- Most columns have object data type, while only release_year is of integer type, suitable for time-based analysis.
- Some columns like director, cast, country, rating, and duration have missing values, indicating data cleaning requirements.
- The dataset is memory-efficient (around 826 KB) and well-structured, making it suitable for exploratory data analysis, visualization, and insight generation.

HANDLING MISSING VALUES

```
nf.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0

dtype: int64

1. **The dataset shows missing values mainly in director (2634), cast (825), and country (831) columns, indicating incomplete metadata for several titles.**
2. **Core columns such as show_id, type, title, release_year, listed_in, and description have no missing values, making them reliable for primary analysis and visualization.**

```
nf['director'] = nf['director'].fillna("Unknown")
nf['cast'] = nf['cast'].fillna("Not Available")
nf['country'] = nf['country'].fillna("Unknown")
nf['date_added'] = nf['date_added'].fillna(method='ffill')
nf['rating'] = nf['rating'].fillna("Not Rated")
nf["duration"] = nf["duration"].fillna("Not Available")
nf.isnull().sum()
```

	0
show_id	0
type	0
title	0
director	0
cast	0
country	0
date_added	0
release_year	0
rating	0
duration	0
listed_in	0
description	0

dtype: int64

- The fillna() operations handle missing values by replacing director and country with “Unknown” and cast and duration with “Not Available”, ensuring consistency.
- The date_added column is filled using forward fill (ffill), preserving the sequence of dates logically.
- Missing values in the rating column are replaced with “Not Rated”, maintaining categorical clarity.
- The final output shows zero null values in all columns, confirming successful data cleaning and readiness for further analysis and visualization.

Descriptive Statistics

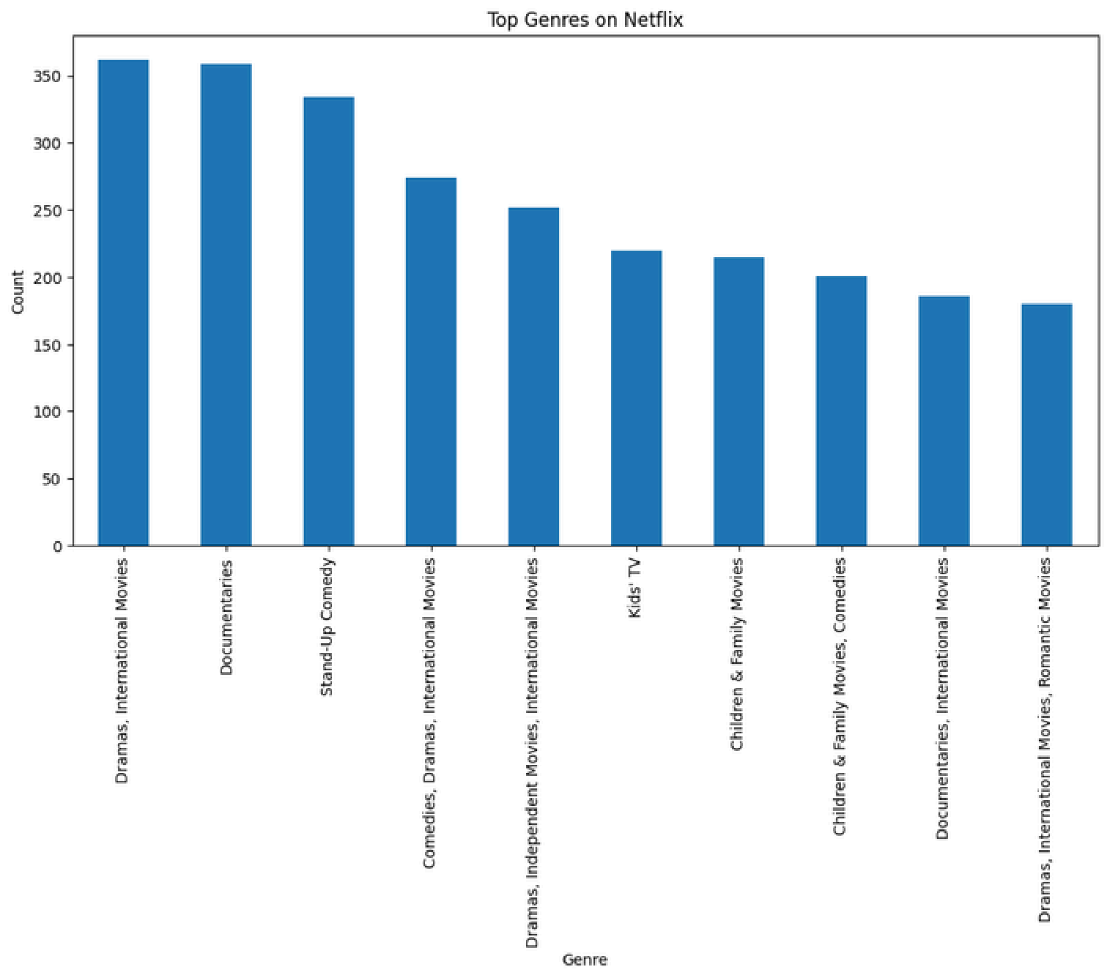
```
nf.describe()
```

	release_year
count	8807.000000
mean	2014.180198
std	8.819312
min	1925.000000
25%	2013.000000
50%	2017.000000
75%	2019.000000
max	2021.000000

- The dataset contains 8,807 titles, indicating complete availability of release year data.
- The average release year is around 2014, showing Netflix's focus on relatively modern content.
- The standard deviation (~8.8 years) suggests moderate variation in content release timelines.
- The median year is 2017, meaning half of the content was released after this year.
- The range spans from 1925 to 2021, highlighting a mix of classic and recent titles, though newer content dominates the platform.

Data Visualization

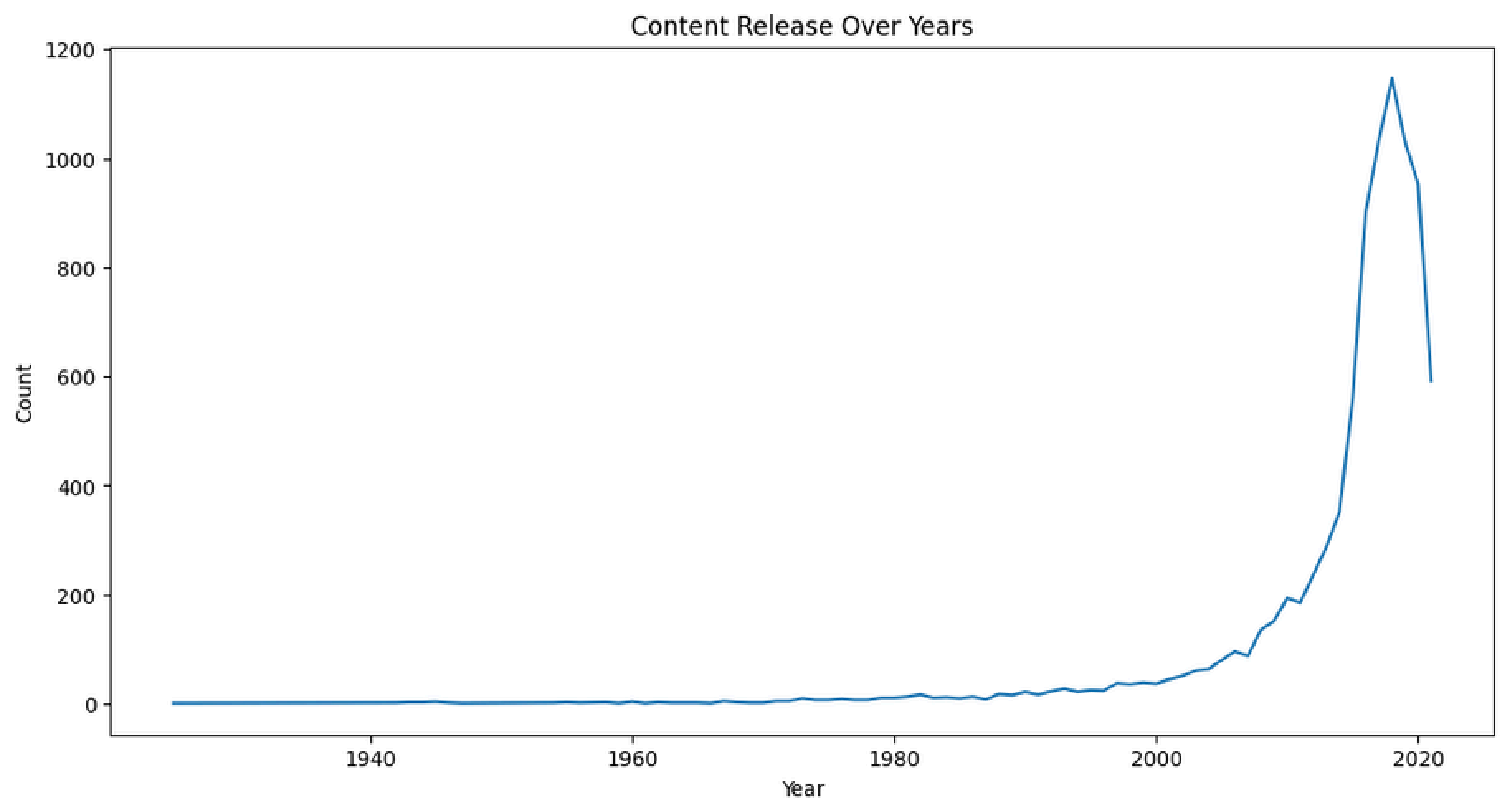
```
plt.figure(figsize=(12,6))
nf['listed_in'].value_counts().head(10).plot(kind='bar')
plt.title("Top Genres on Netflix")
plt.xlabel("Genre")
plt.ylabel("Count")
plt.show()
```



- **Graph clearly shows that Dramas (especially International Movies) and Documentaries are the most dominant genres on Netflix, indicating high content availability in these categories.**
- **The strong presence of Stand-Up Comedy and Children/Kids content reflects Netflix’s focus on both entertainment-driven and family-friendly audiences.**

Visualize the distribution of content across release years.

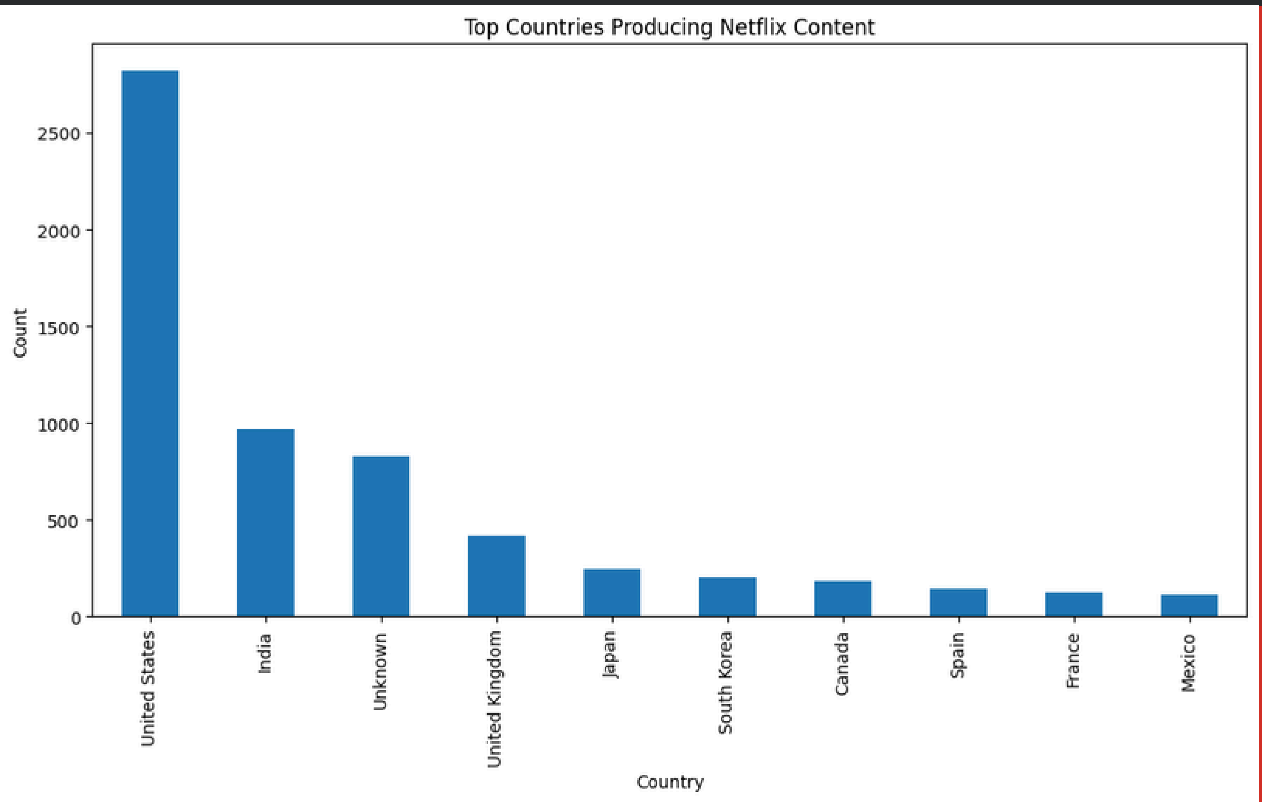
```
plt.figure(figsize=(12,6))
nf['release_year'].value_counts().sort_index().plot(kind='line')
plt.title("Content Release Over Years")
plt.xlabel("Year")
plt.ylabel("Count")
plt.show()
```



- **The code analyzes content count by release_year, showing how the number of Netflix releases has changed over time.**
- **The graph clearly indicates a sharp increase in content releases after 2015, highlighting Netflix's rapid expansion and strong focus on producing and acquiring new content in recent years.**

Geographical Distribution of Content

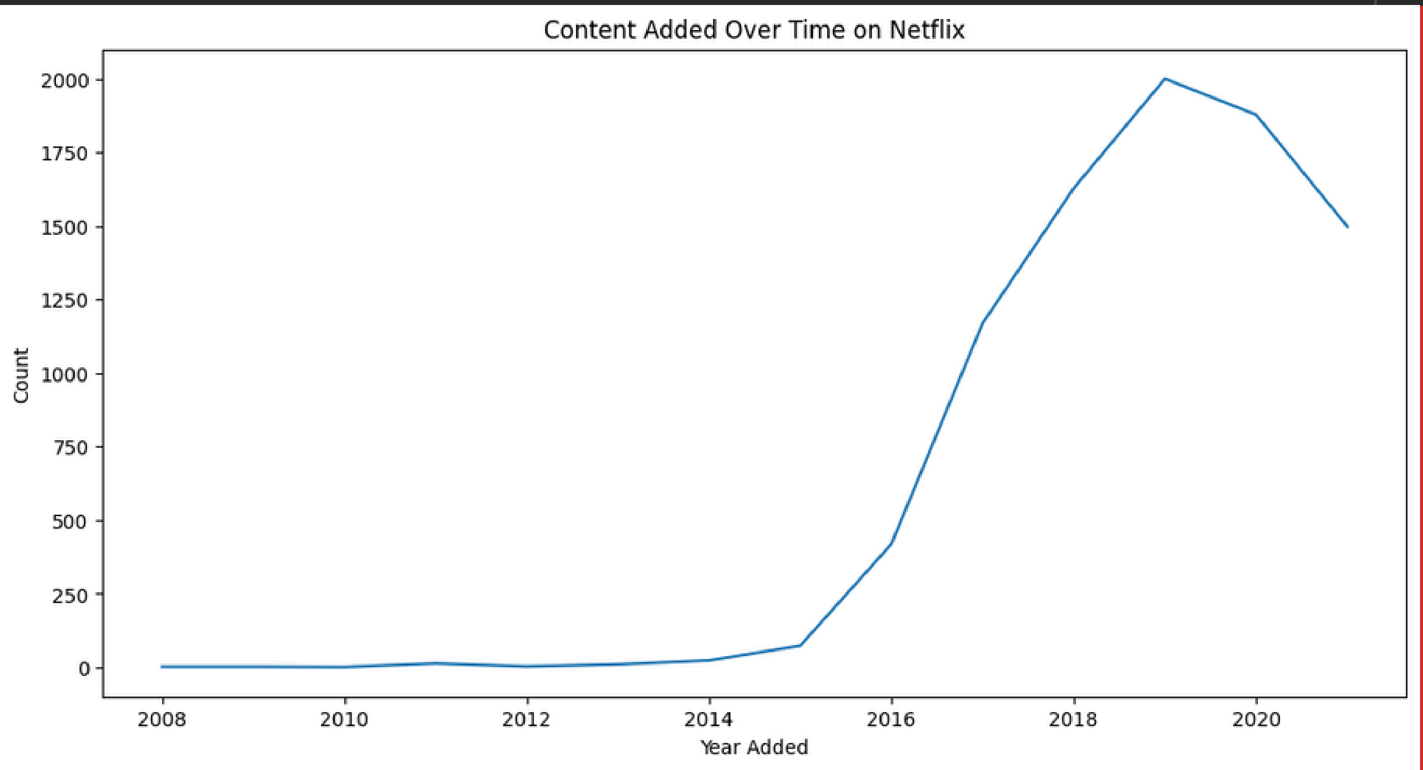
```
plt.figure(figsize=(12,6))
nf['country'].value_counts().head(10).plot(kind='bar')
plt.title("Top Countries Producing Netflix Content")
plt.xlabel("Country")
plt.ylabel("Count")
plt.show()
```



- **The code uses country.value_counts() to identify the top 10 countries producing Netflix content, visualized through a bar chart.**
- **The graph shows the United States as the largest content producer, followed by India and the United Kingdom, highlighting Netflix's strong presence in major global markets.**

Time Series Analysis

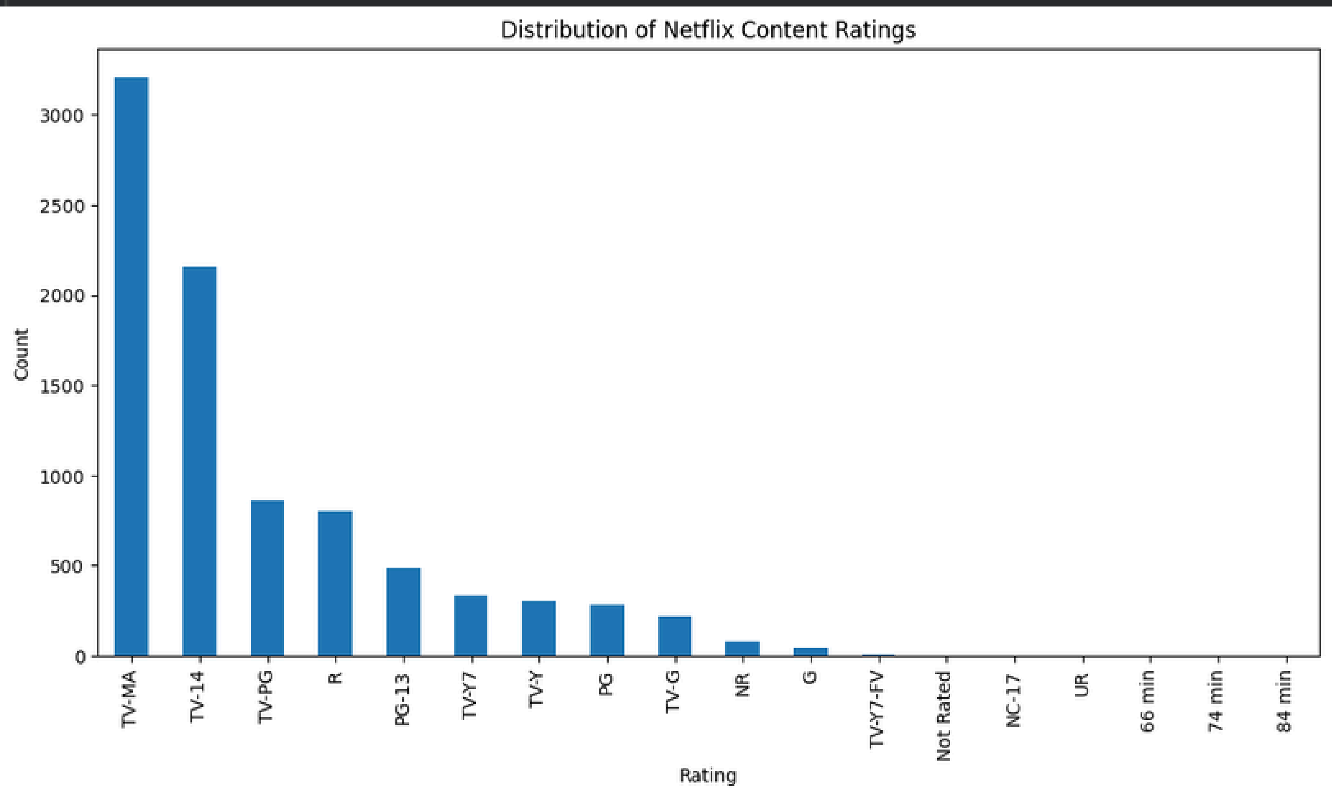
```
nf['date_added'] = pd.to_datetime(nf['date_added'], format='%B %d, %Y', errors='coerce')
nf['year_added'] = nf['date_added'].dt.year
nf['year_added'].value_counts().sort_index().plot(figsize=(12,6))
plt.title("Content Added Over Time on Netflix")
plt.xlabel("Year Added")
plt.ylabel("Count")
plt.show()
```



1. **The code converts date_added into datetime and extracts year_added.**
2. **The graph shows very low content additions before 2015.**
3. **A sharp rise starts from 2016 onwards.**
4. **Content addition peaks around 2019.**
5. **Slight decline after 2020 indicates slowed expansion.**

Distribution of Content Ratings

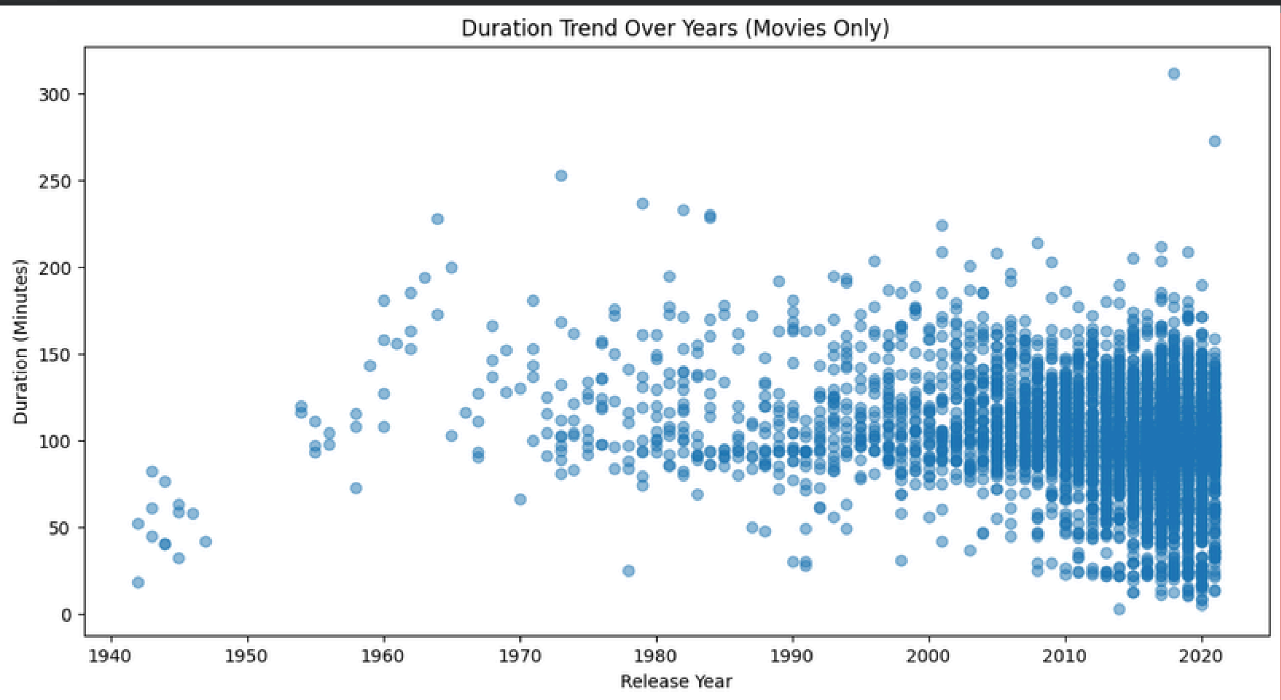
```python
plt.figure(figsize=(12,6))
nf['rating'].value_counts().plot(kind='bar')
plt.title("Distribution of Netflix Content Ratings")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.show()
```



1. **The code uses value_counts() on the rating column to show the frequency of each rating.**
2. **The graph shows that TV-MA is the most common rating, meaning Netflix has more adult-oriented content.**
3. **TV-14 and TV-PG are next, indicating a strong presence of teen-friendly content.**
4. **Ratings like G and NC-17 are very rare, meaning strictly-family or highly-restricted content is limited**

Explore The Length of Movies or Episodes and Identify Any Trends

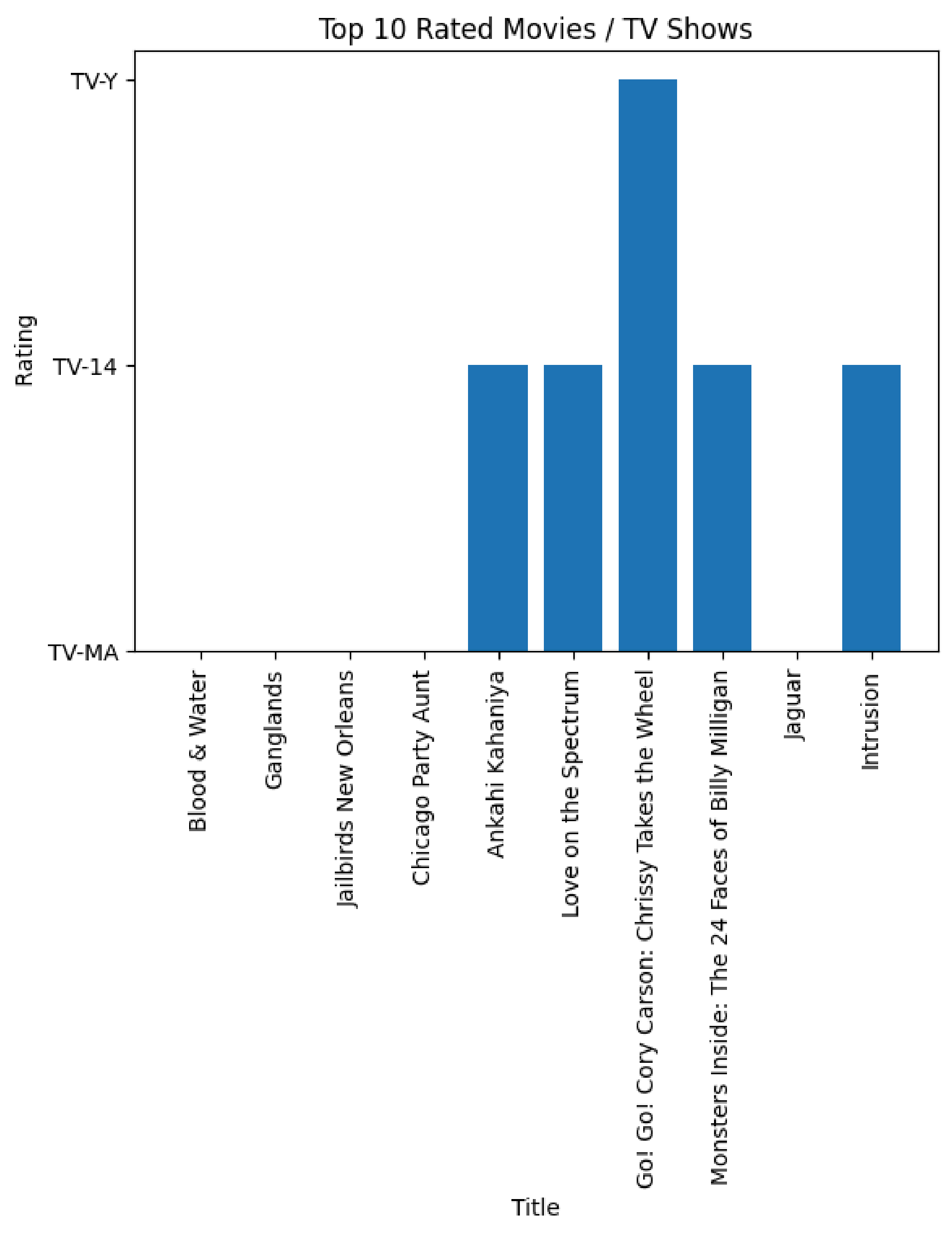
```
movies_df = nf[nf['type'] == 'Movie'].copy()
# Filter out rows where 'duration' is 'Not Available' before conversion
movies_df = movies_df[movies_df['duration'] != 'Not Available'].copy()
movies_df['duration_int'] = movies_df['duration'].str.replace(' min', '').astype(int)
plt.figure(figsize=(12,6))
plt.scatter(movies_df['release_year'], movies_df['duration_int'], alpha=0.5)
plt.title("Duration Trend Over Years (Movies Only)")
plt.xlabel("Release Year")
plt.ylabel("Duration (Minutes)")
plt.show()
```



1. The code filters only Movies, converts duration into minutes, and plots Release Year vs Duration as a scatter plot.
2. The graph shows that movie durations vary widely, but in recent years most Netflix movies cluster between 80–130 minutes, indicating a trend toward standard-length films.

Top-Rated Movies or TV Shows Based On User Ratings.

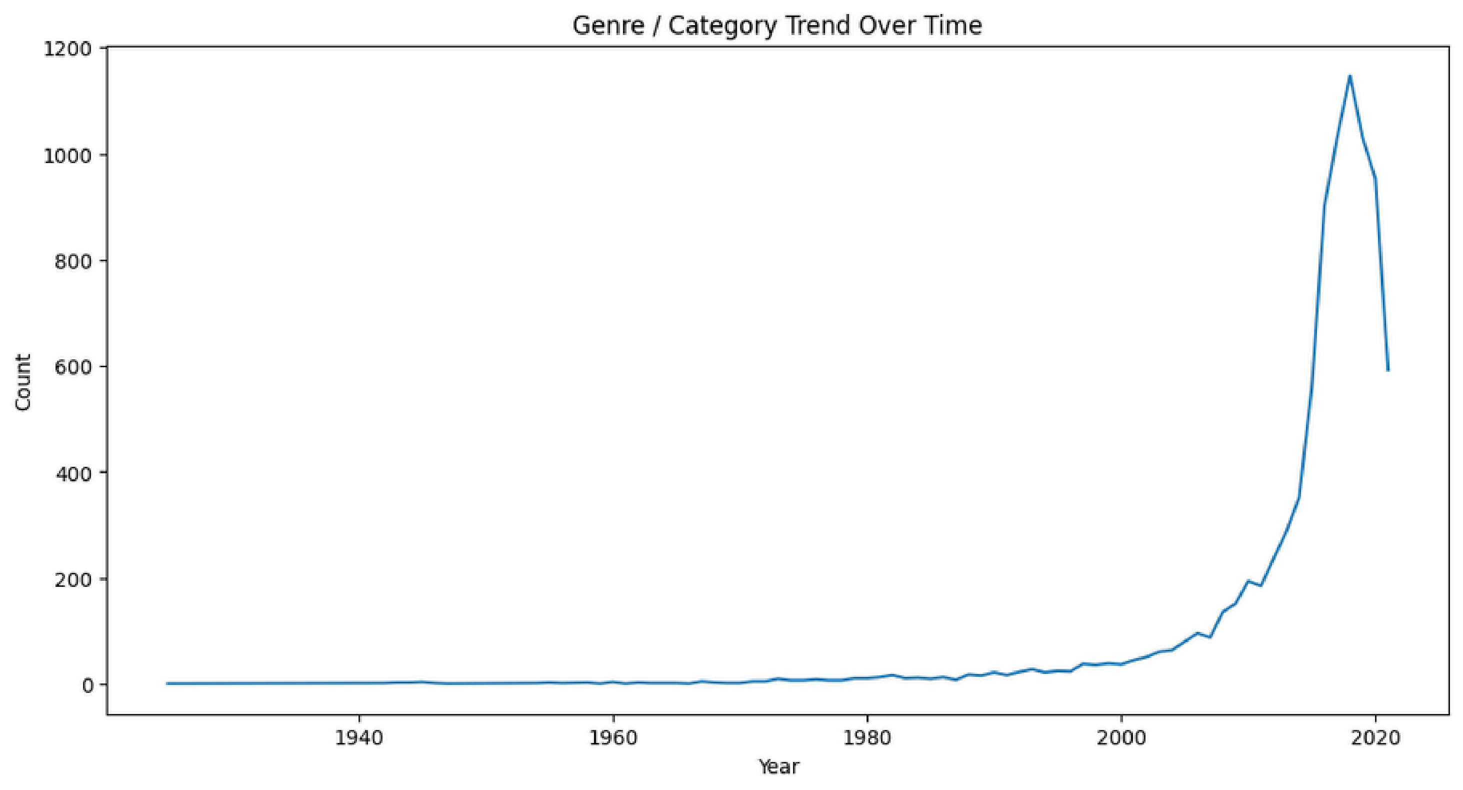
```
plt.figure()
plt.bar(top_rated['title'], top_rated['rating'])
plt.xticks(rotation=90)
plt.xlabel('Title')
plt.ylabel(' Rating')
plt.title('Top 10 Rated Movies / TV Shows')
plt.show()
```



- The code creates a bar chart of the Top-10 rated Netflix titles, showing each title on the x-axis and its rating category on the y-axis. The graph highlights that most top-rated titles fall under TV-MA and TV-14, meaning the highest-rated popular shows and movies are mainly for mature and teenage audiences

Genre / Category Trend Over Time

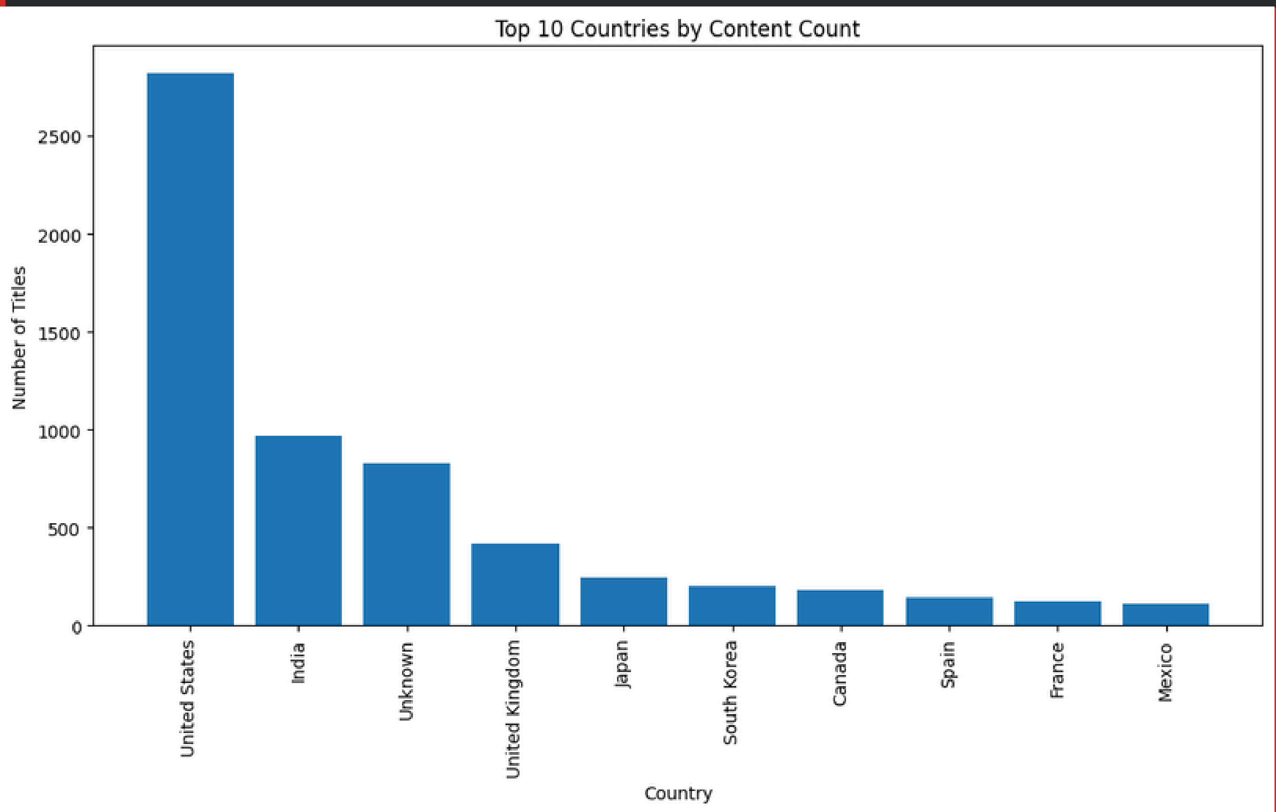
```
genre_year = nf.groupby(['release_year'])['listed_in'].count()
plt.figure(figsize=(12,6))
genre_year.plot()
plt.title("Genre / Category Trend Over Time")
plt.xlabel("Year")
plt.ylabel("Count")
plt.show()
```



The code groups Netflix titles by release year and counts how many titles appear in each genre category per year, then plots the trend. The graph shows very little content before 2000, followed by a sharp and continuous rise after 2010, indicating a rapid expansion of Netflix content production and availability in recent years.

Geographical Analysis

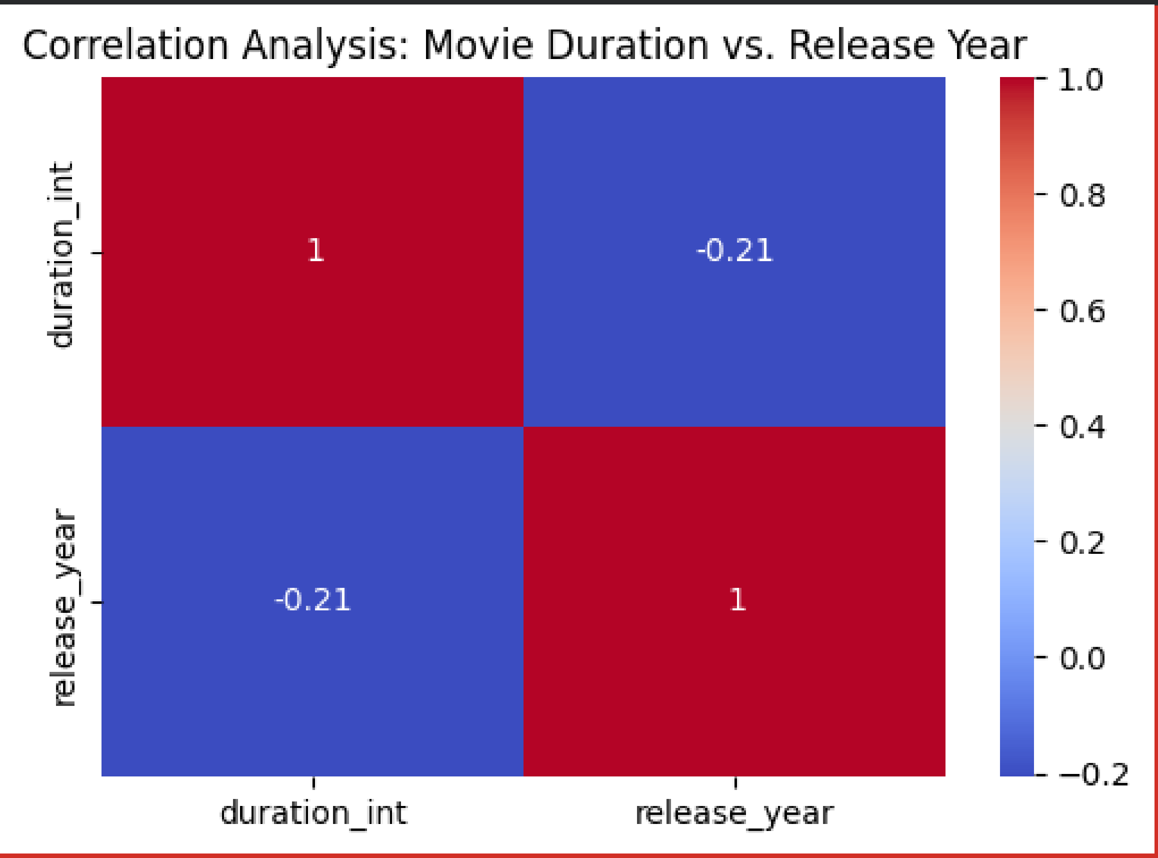
```
country_counts = nf['country'].value_counts().head(10)
plt.figure(figsize=(12,6))
plt.bar(country_counts.index, country_counts.values)
plt.xticks(rotation=90)
plt.xlabel('Country')
plt.ylabel('Number of Titles')
plt.title('Top 10 Countries by Content Count')
plt.show()
```



1. The code counts how many Netflix titles come from each country and selects the Top-10 countries, then plots them in a bar chart.
2. The chart shows the United States dominates by a large margin, followed by India and the UK, meaning most Netflix content is produced in these countries

Correlation Analysis

```
plt.figure(figsize=(6,4))
numeric_nf = movies_df[['duration_int', 'release_year']]
sns.heatmap(numeric_nf.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Analysis: Movie Duration vs. Release Year")
plt.show()
```



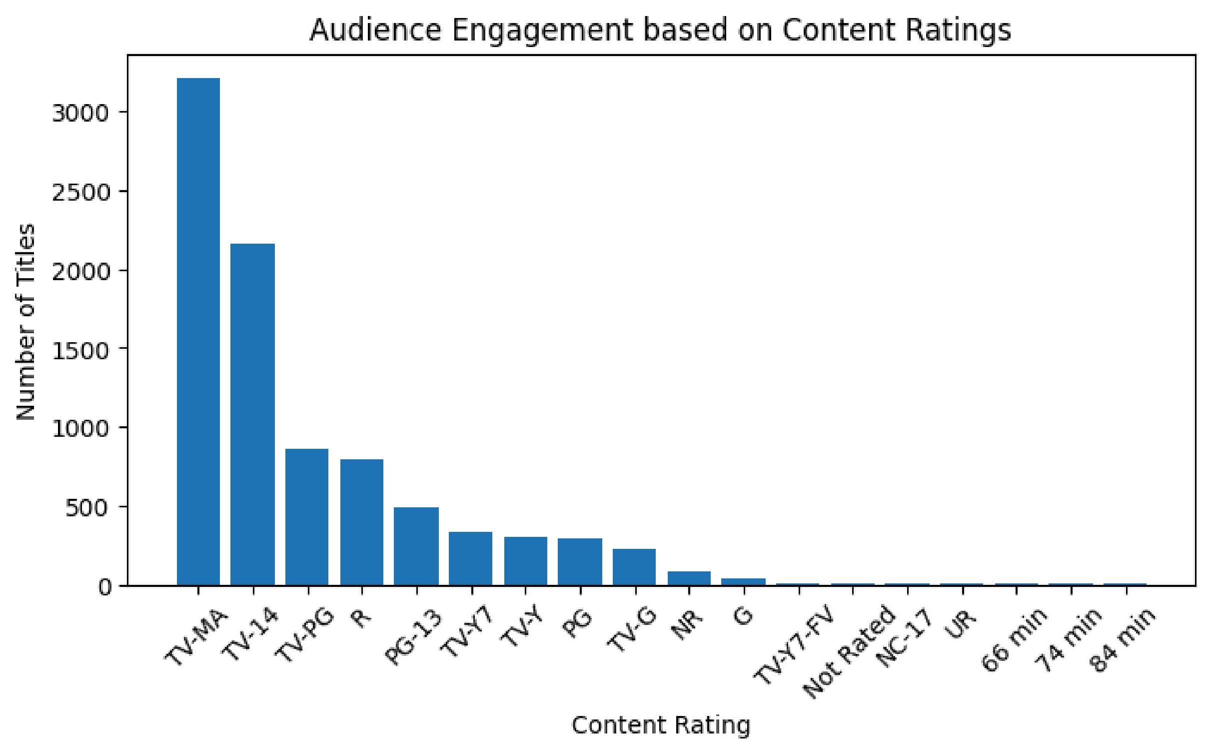
The code selects movie duration and release year, then calculates their correlation and visualizes it using a heatmap.

The correlation value is -0.21, which indicates a weak negative relationship.

Most important: Newer movies tend to be slightly shorter than older movies, but the effect is not very strong.

Audience Engagement

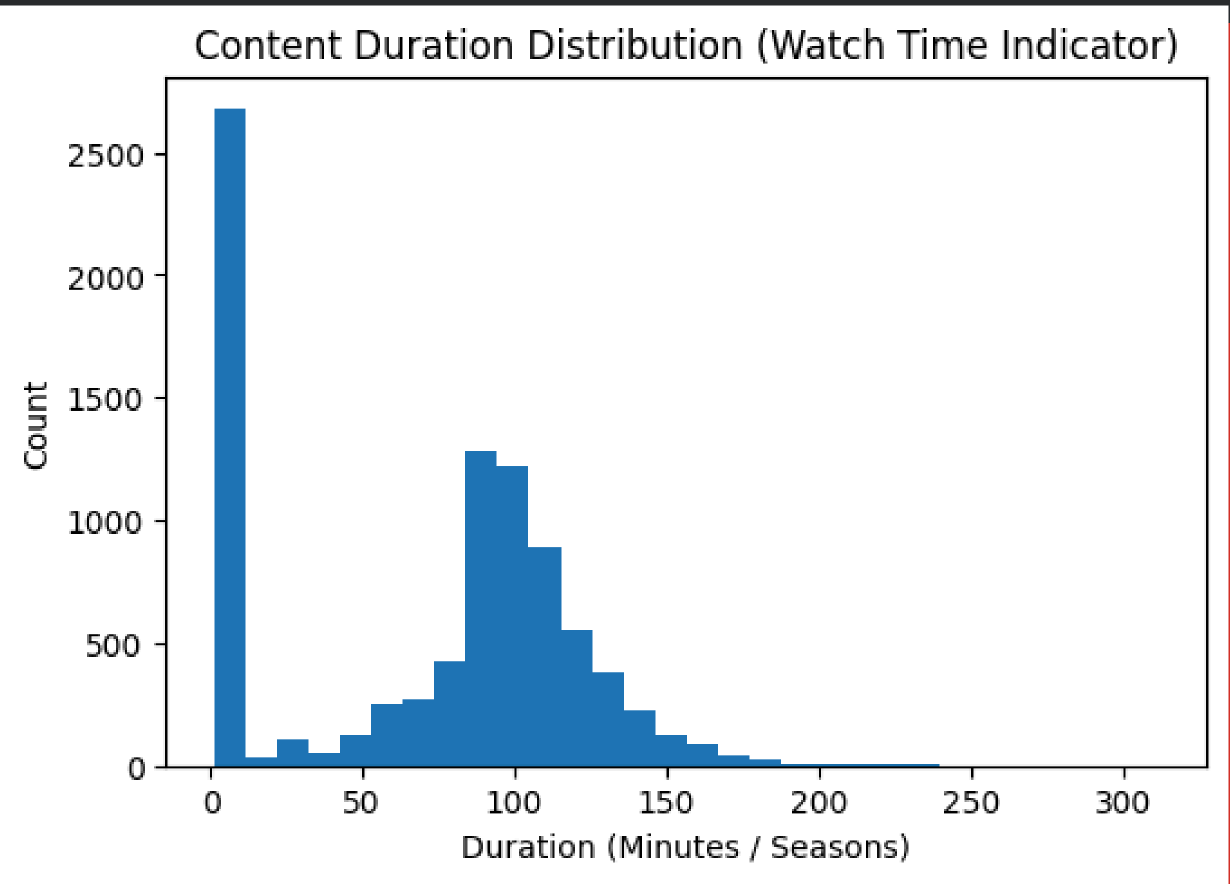
```
rating_counts = nf['rating'].value_counts()
plt.figure(figsize=(8,4))
plt.bar(rating_counts.index, rating_counts.values)
plt.title("Audience Engagement based on Content Ratings")
plt.xlabel("Content Rating")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```



1. **The code counts titles by content rating and plots them in a bar chart.**
2. **TV-MA has the highest number of titles, followed by TV-14 and TV-PG, meaning mature and teen content dominates**

Content Duration Distribution (Watch Time Indicator)

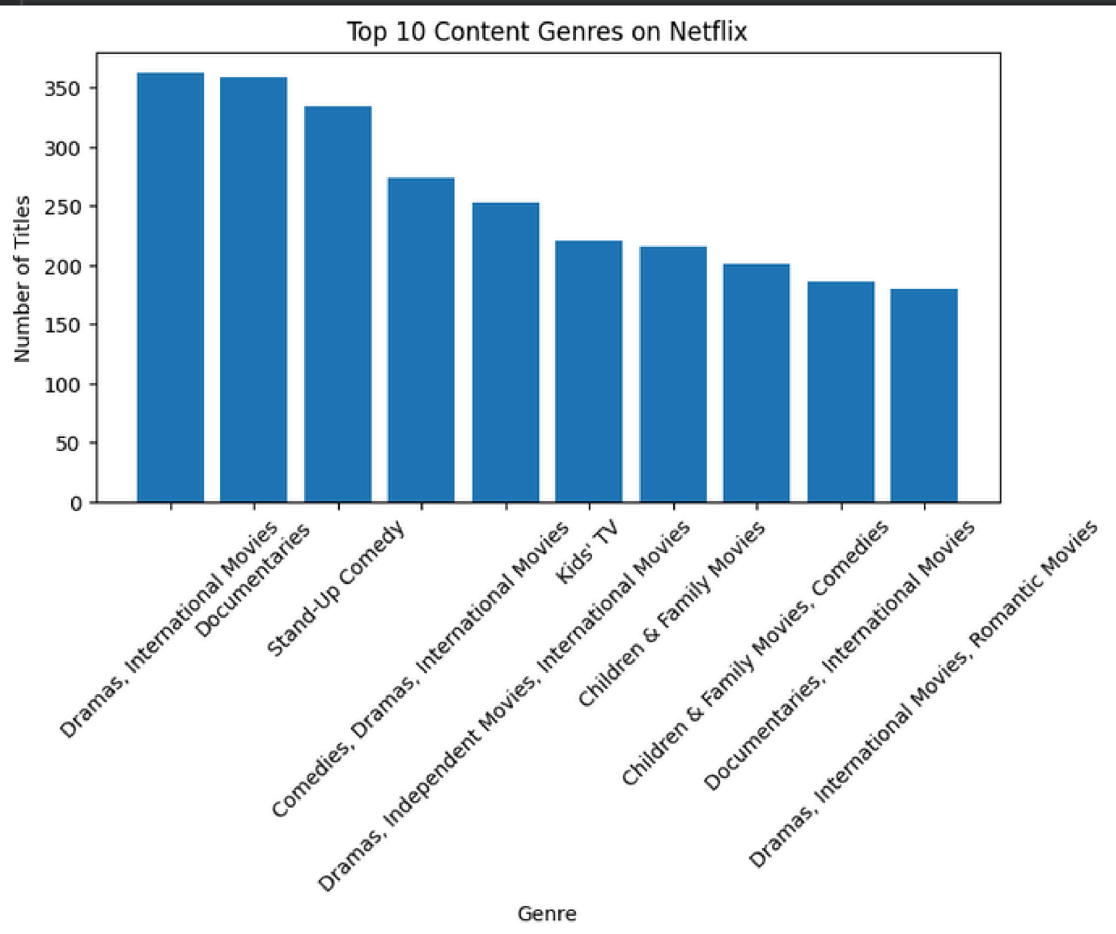
```
nf['duration_int'] = nf['duration'].str.extract('(\d+)').astype(float)
plt.figure(figsize=(6,4))
plt.hist(nf['duration_int'].dropna(), bins=30)
plt.title("Content Duration Distribution (Watch Time Indicator)")
plt.xlabel("Duration (Minutes / Seasons)")
plt.ylabel("Count")
plt.show()
```



1. **The code extracts numeric values from the duration column and converts them into a numeric variable for analysis.**
2. **A histogram is plotted to show the overall distribution of watch duration.**
3. **Most Netflix titles fall between 60–120 minutes, while very long durations (200+ minutes) are rare.**

Content Variety

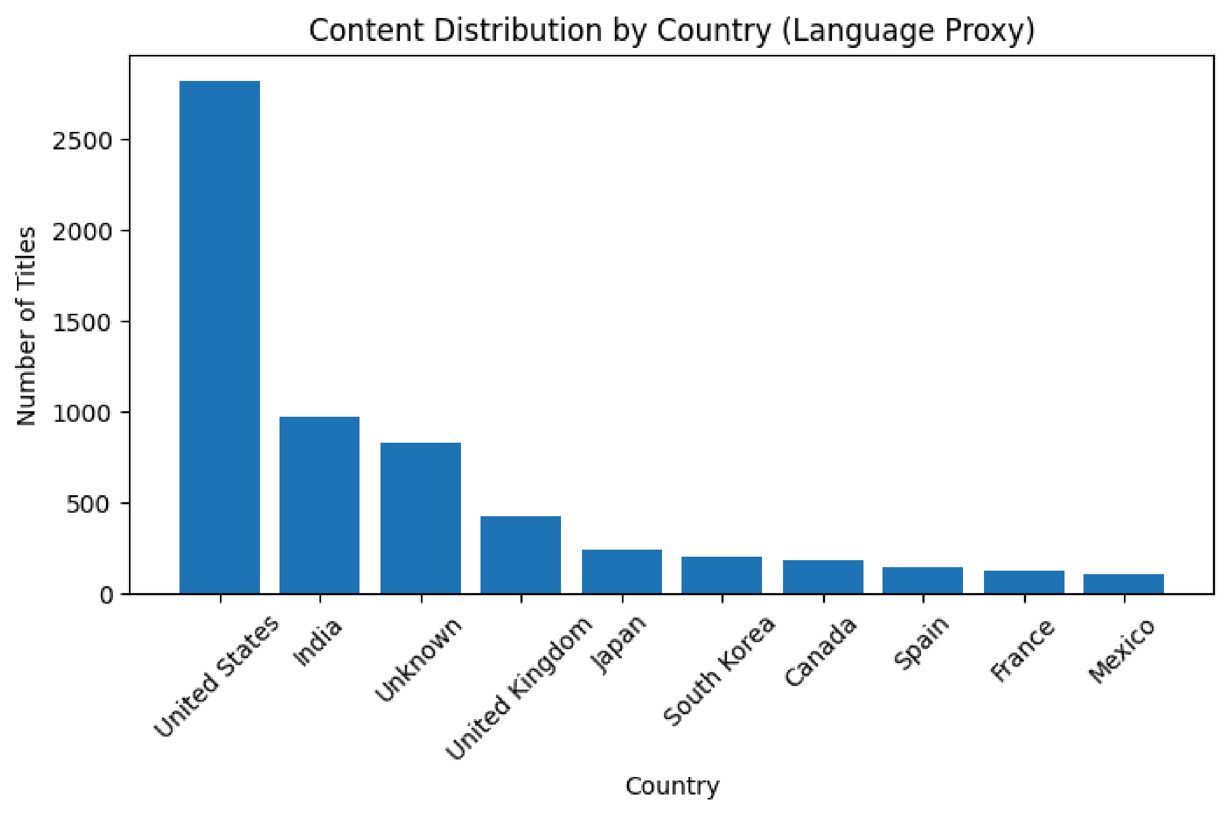
```
genre_counts = nf['listed_in'].value_counts().head(10)
plt.figure(figsize=(8,4))
plt.bar(genre_counts.index, genre_counts.values)
plt.title("Top 10 Content Genres on Netflix")
plt.xlabel("Genre")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```



1. The code counts how often each genre/category appears in the listed_in column and selects the Top-10 genres, then visualizes them using a bar chart.
2. The graph shows Drama and International Movies dominate Netflix’s library, followed by Comedy and Documentary content, meaning Netflix strongly focuses on story-driven and global content.

Language Analysis

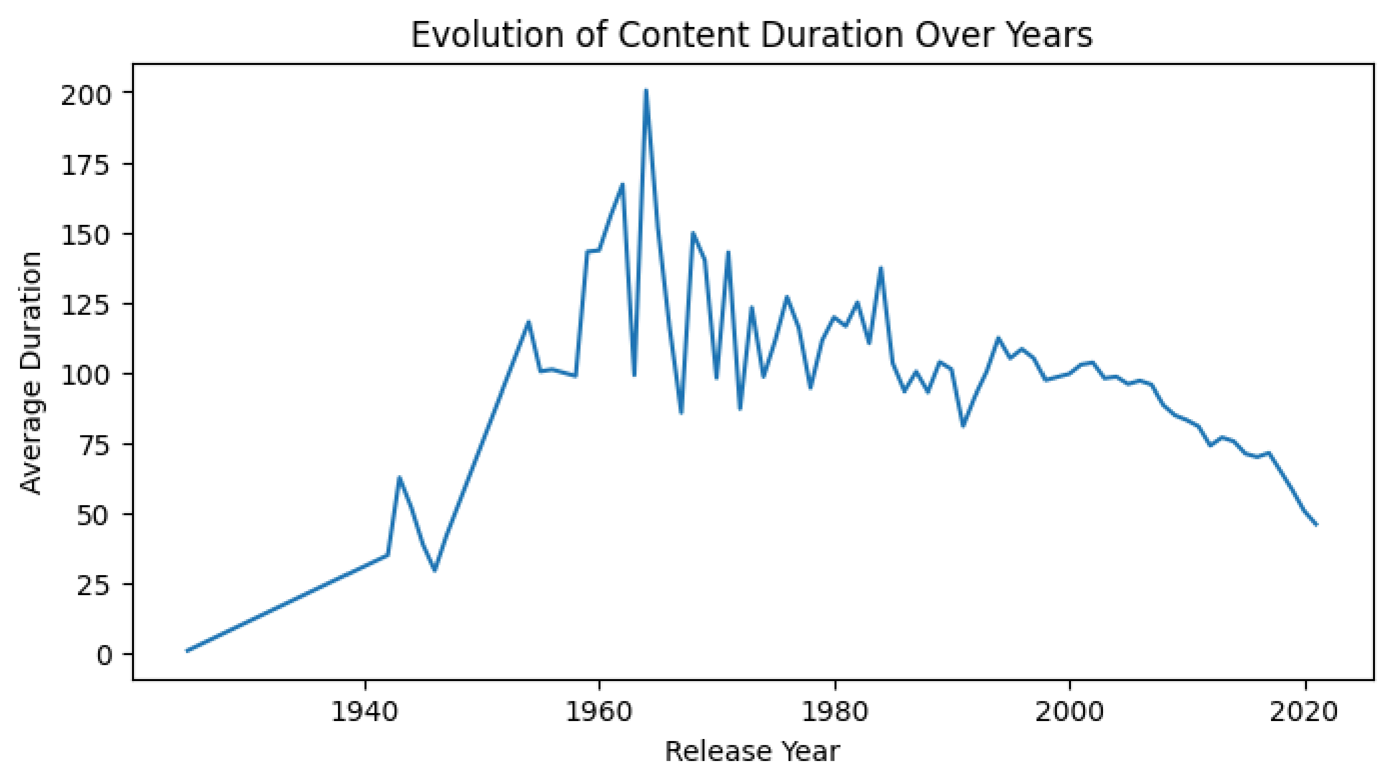
```
plt.figure(figsize=(8,4))
plt.bar(country_counts.index, country_counts.values)
plt.title("Content Distribution by Country (Language Proxy)")
plt.xlabel("Country")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)
plt.show()
```



1. **The code uses the country field to count how many Netflix titles originate from each country and displays the distribution with a bar chart. The graph shows that the United States contributes the largest share of content, followed by India and the United Kingdom. Other countries like Japan, South Korea, Canada, Spain, France, and Mexico contribute smaller but noticeable amounts.**

Evolution of Content Duration Over Years

```
duration_year = nf.groupby('release_year')['duration_int'].mean()
plt.figure(figsize=(8,4))
plt.plot(duration_year.index, duration_year.values)
plt.title("Evolution of Content Duration Over Years")
plt.xlabel("Release Year")
plt.ylabel("Average Duration")
plt.show()
```

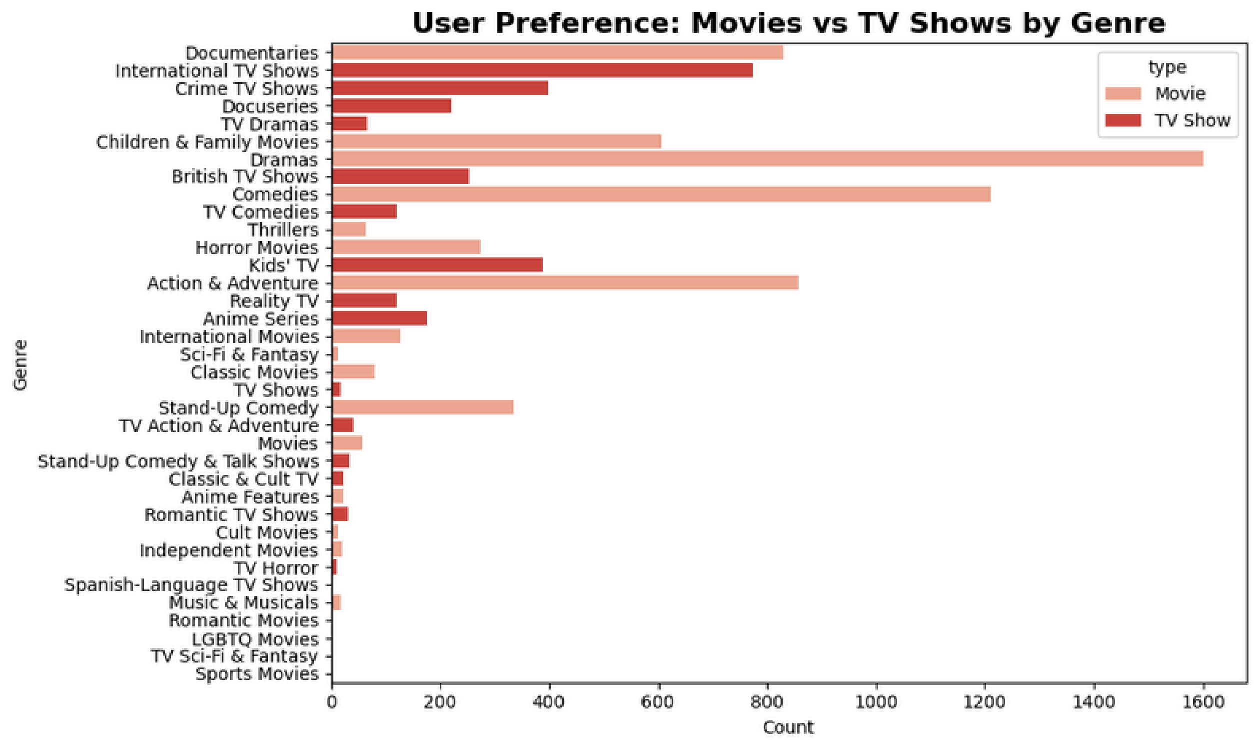


1. **The code groups Netflix titles by release year and calculates the average duration for each year, then plots a line chart. The graph shows movie duration peaked around the 1960s, then gradually declined. In recent years, the average duration has reduced significantly, suggesting modern content is becoming shorter.**

User Preferences

```
nf["main_genre"] = nf["listed_in"].str.split(",").str[0]
```

```
plt.figure(figsize=(10,6)),
sns.countplot(data=nf, y='main_genre', hue="type", palette="Reds")
plt.title("User Preference: Movies vs TV Shows by Genre",fontsize=16,fontweight=
          'bold')
plt.xlabel("Count")
plt.ylabel("Genre")
plt.tight_layout()
plt.show()
```



- **The code creates a countplot comparing Movies vs TV Shows across different genres using the main_genre column. The graph shows that Documentaries, Dramas, and International TV Shows are highly popular, with TV Shows dominating many genres. Movies are more frequent in genres like Action & Adventure and Horror, showing different audience preferences.**

REFRENCES

1. **Dataset Source: Skill Circle Restaurant Data for Analysis**
2. **Development Environment: Google Colab Python**
3. **Libraries Used: Pandas, NumPy, Matplotlib**

CONCLUSION

The Netflix dataset provides valuable insights into the platform's diverse and expanding content library. It includes a large number of movies and TV shows with detailed information such as release year, genre, country, rating, and duration. After handling missing values, the dataset becomes clean and reliable for analysis. The statistical summary shows that most content is released after 2010, highlighting Netflix's focus on modern and trending entertainment. The presence of multiple genres and countries reflects global content diversity. Overall, this dataset is highly suitable for exploratory data analysis, visualization, and understanding content trends, audience preferences, and strategic decisions in the streaming industry.


Thank You!!
NETFLIX